

Product Requirements Document (PRD)

Product Name: Mechanical Robotics Portfolio Web Application

Candidate / Owner: Pyla Amruth Varun

Target Delivery Stack: MERN Stack (MongoDB, Express.js, React.js, Node.js)

Status: Ready for Development

1. Executive Summary & Objective

The objective of this web application is to deliver a developer-styled portfolio landing page tailored for a **Mechanical & Robotics Engineer**. It translates hardware, CAD, kinematic simulations, and industrial automation experience into an interactive web experience. Built on the MERN stack, the application combines a frontend (React + Tailwind CSS) with a backend API (Node/Express + MongoDB) to manage dynamic case studies, handle inquiry submissions, and track portfolio analytics.

2. Target Audience & User Personas

- **Robotics / Hardware Engineering Recruiters:** Looking for fast verification of core mechanical CAD tools (CATIA V5, SolidWorks, AutoCAD), kinematic validation, and manufacturing experience (CNC, 3D printing).
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- **Automation Engineering Managers:** Seeking technical depth in mechanisms, motion simulation, and DFM (Design for Manufacturing).
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- **Collaborators & Industry Clients:** Evaluating technical capabilities for robotics design, automation systems, and CAD prototyping contracts.
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3. Tech Stack Architecture

- **Frontend:** React.js, Tailwind CSS, Lucide React (Icons), Framer Motion (Animations, Terminal Typing Effects).

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- **Backend:** Node.js, Express.js (REST API, input validation, rate limiting).
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- **Database:** MongoDB Atlas with Mongoose ODM (Data modeling for projects, skills, contacts).
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- **Communication & Delivery:** Nodemailer / Brevo API (Automated email notifications for contact forms).
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- **Hosting & CDN:** Vercel / Netlify (Client), Render / Railway (Backend API), Cloudinary (CAD media & render asset storage).
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4. System Features & Functional Specifications

4.1 Client-Facing Features (Frontend)

- **Sticky Terminal Navbar:**
- - Displays monospace brand tag `varun://robotics`.
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 - Anchor navigation (`// Stack`, `// Projects`, `// Experience`, `// Contact`).
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 - Direct action triggers: `Download Resume` (served via static/CDN endpoint) and `Let's Talk` CTA.
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- **Interactive Hero Section:**
- - Headline: *"Engineering Hardware, Kinematics & Automation Systems"*.
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 - Dynamic Code Terminal Card with simulated typewriter output displaying live status, skills, and current role at DJT EV POWERTECH.
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 - Quick-action buttons linking to GitHub, LinkedIn, and the project showcase.
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- **Categorized Skills Bento Matrix:**
- - Dynamic rendering of tools grouped into: *CAD & Mechanical Design*, *Robotics & Simulation*, *Manufacturing & Prototyping*, and *Software & Embedded*.
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- **Featured Projects Showcase:**
- - Multi-card grid populated from MongoDB with dynamic filter tags (`All`, `Robotics`, `CAD/DFM`, `Automation`).
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- Core seed projects:
 - 1. *3DOF RPR Robot Kinematic Analysis* (MATLAB Simulink, RoboAnalyzer)
 - 2.
 3. *EV Charging Station Enclosure Design* (CATIA V5, AutoCAD)
 - 4.
 5. *Collaborative Robot (Cobot) Structure* (CATIA V5)
 - 6.
 7. *Articulating Laparoscopic Surgical Tool* (CATIA V5, 3D Printing)
 - 8.
 9. *Drone Frame Assembly with Betaflight Avionics* (Betaflight, CAD)
 - 10.
- **Industrial Experience Timeline:**
 - - Chronological expandable tree for DJT EV POWERTECH, Hunumax Technologies, and Dreambots.
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- **Contact & Inquiry Module:**
 - - Direct contact details (amruthvarunpyla@gmail.com, [+91 9052926058](tel:+919052926058)) with one-click clipboard copy.
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 - Contact form: Name, Email, Subject, Message, and Recruiter/Company toggle.
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4.2 Backend & API Specifications (Node.js + Express)

API Endpoints Table

Method	Endpoint	Description	Auth Required
GET	/api/v1/projects	Fetch all project records (supports ?category= filter)	No
GET	/api/v1/projects/:id	Fetch detailed single project breakdown	No

POST	/api/v1/projects	Create a new project entry	Yes (JWT)
PUT	/api/v1/projects/:id	Update an existing project	Yes (JWT)
DELETE	/api/v1/projects/:id	Delete a project	Yes (JWT)
POST	/api/v1/contact	Submit contact inquiry form & trigger email alert	No (Rate-Limited)
POST	/api/v1/admin/login	Authenticate admin user & issue JWT	No

5. Database Schema Designs (MongoDB / Mongoose)

5.1 Project Schema (Project.js)

JavaScript

```
const mongoose = require('mongoose')

const Project = mongoose.model('Project', new mongoose.Schema({
  title: { type: String, required: true },
  slug: { type: String, required: true, unique: true },
  category: {
    type: String,
    enum: ['Robotics', 'CAD/DFM', 'Automation', 'Biomedical', 'Avionics'],
    required: true
  },
  shortDescription: { type: String, required: true },
  fullDescription: { type: String },
  toolsUsed: { type: String },
  keyOutcomes: { type: String },
  media: {
    thumbnailUrl: { type: String, required: true },
    cadRenders: { type: String },
    videoDemoUrl: { type: String }
  },
  links: {
    github: { type: String },
    cadLibrary: { type: String },
    liveDemo: { type: String }
  }
})
})
```

```
featured type Boolean default false
orderIndex type Number default 0
timestamps true
```

```
module 'Project'
```

5.2 Contact Inquiry Schema (Inquiry.js)

JavaScript

```
const require 'mongoose'
```

```
const new
name type String required true trim true
email type String required true trim true lowercase true
subject type String default 'General Inquiry'
message type String required true
ipAddress type String
status type String enum 'UNREAD' 'READ' 'RESPONDED' default 'UNREAD'
timestamps true
```

```
module 'Inquiry'
```

6. Non-Functional Requirements (NFRs)

- **Performance:** Google Lighthouse score > 90 across Performance, Accessibility, Best Practices, and SEO. Initial First Contentful Paint (FCP) < 1.2s.
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- **Security:**
- - Rate-limiting on `/api/v1/contact` (max 5 requests per 15 minutes per IP using `express-rate-limit`).
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 - Cross-Origin Resource Sharing (CORS) restricted to verified frontend domains.
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 - Sanitization of form inputs using `express-validator` to prevent XSS and NoSQL injection.
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- **Responsiveness:** Full layout adaptation across Mobile (<640px), Tablet (640px–1024px), and Desktop (>1024px).
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- **Asset Optimization:** All 3D CAD renders converted to modern WebP format with lazy loading enabled.
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7. Implementation Roadmap & Milestones

- **Sprint 1 — Backend Infrastructure & Schemas (Days 1–2):** Set up Node.js/Express server, MongoDB Atlas cluster connection, Mongoose schemas, seed data script with resume projects, and `/api/v1/projects` endpoints.

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- **Sprint 2 — Frontend Core Layout & UI System (Days 3–4):** Initialize React (Vite) + Tailwind CSS, implement color theme, fonts, Navbar, Hero section, and Terminal card.
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- **Sprint 3 — Project Grid & Experience Module (Days 5–6):** Connect React frontend to Express API via Axios/React Query, build responsive Bento project grid, dynamic filters, and modal view for CAD assets.
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- **Sprint 4 — Contact Form, Email Delivery & Polishing (Days 7–8):** Build contact form with client validation, Nodemailer integration on backend, Framer Motion micro-interactions, responsive testing, and Vercel/Render deployment.